

Long-range order and spontaneous symmetry breaking in quantum spin systems and hard-core bosons on a lattice

part 1 introduction and notation

***Advanced Topics in
Statistical Physics
by Hal Tasaki***

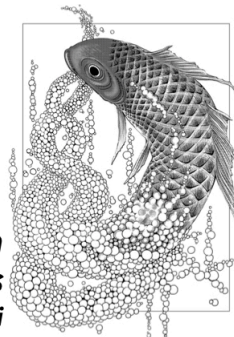


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exact ground state with LRO but without SSB
vs

macroscopic ground state with full SSB

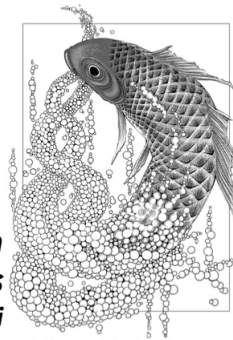
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quantum many-body systems in which the Hamiltonian and the order operator do NOT commute

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answers common questions regarding Bogoliubov's theory of BEC and the BCS theory

Q: do these theories indicate BEC/superconductivity is suppressed $\rightarrow$ NO
when the particle number is fixed?

Q: do they mean BEC/superconductivity inevitably has large particle number fluctuation? $\rightarrow$ NO, not in a single system
yes, for coupled systems

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§ introduction

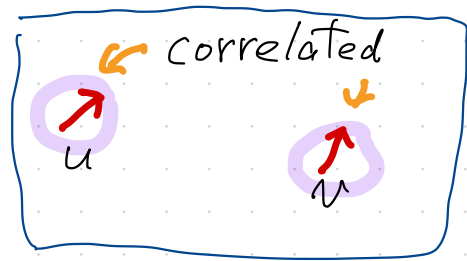
characterization of ordered phases in macroscopic physical systems
crystalline order, (anti)ferromagnetic order
(particle condensation (BEC, superconductivity))

→ NOT modern topological phases

▶ long-range order (LRO)

$$\langle \hat{S}_u \cdot \hat{S}_v \rangle \simeq \text{const} > 0 \quad \text{as } |u-v| \rightarrow \infty$$

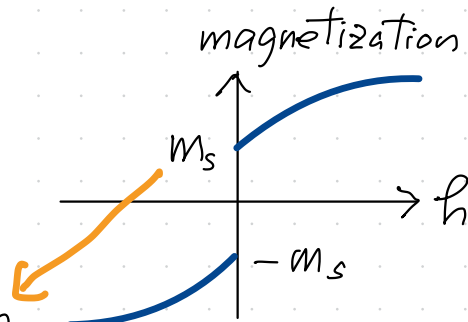
elements far apart are correlated



▶ Spontaneous symmetry breaking (SSB)

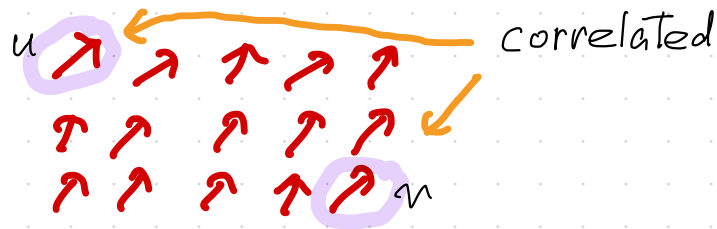
realized states (in the infinite system)
break the symmetry of the Hamiltonian

nonzero spontaneous magnetization



the relations between LRO and SSB

a state with SSB always has LRO



states with LRO without SSB ? → even experimentally! (part 5)

possible and often appears mathematically/numerically

an equilibrium/ground state with LRO without SSB

⇓ Griffiths-type theorem

$\Rightarrow$ equilibrium/ground states with SSB (in the infinite volume limit)

a ground state with LRO without SSB $\Rightarrow$ low-lying excited states

(in a finite volume)

Which states are physically realistic?

3

parts 3 and 4

part 5

states exhibiting —

quantum spin systems

particle condensation

LRO
and
full SSB

physical

appear in many
theoretical treatments

LRO
and
no SSB

"observed"
theoretically and
numerically

physical !!



LRO should be ODLRO

(off-diagonal long-range order)

part 1 introduction and notation

part 2 LRO and SSB in the equilibrium states of the classical Ising model

part 3 LRO and SSB in the ground states of quantum spin systems with discrete symmetry

part 4 LRO and SSB in the ground states of the quantum XY model

part 5 ODLRO and SSB in the ground states of hard-core bosons on a lattice

§ lattice and the set of bonds

dimension $d = 1, 2, 3, \dots$

side length $L = 4, 6, 8, \dots$

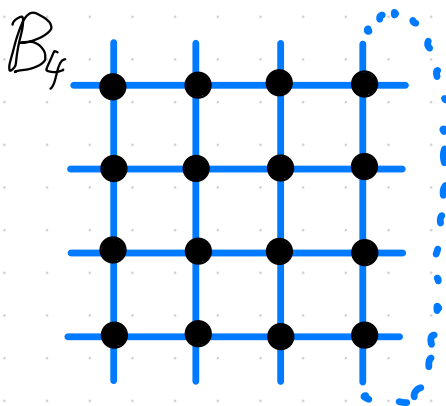
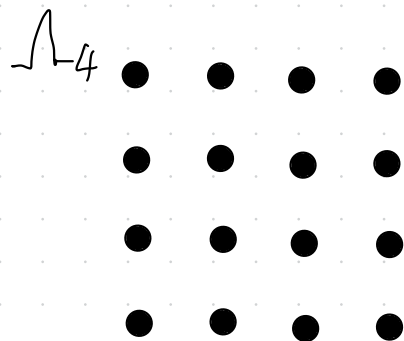
d-dimensional hypercubic lattice

$$\Lambda_L = \left\{ -\frac{L}{2} + 1, \dots, \frac{L}{2} \right\}^d \subset \mathbb{Z}^d \quad (1)$$

sites $u, v, \dots \in \Lambda_L$ $u = (u_1, u_2, \dots, u_d)$ with $u_j \in \{-\frac{L}{2} + 1, \dots, \frac{L}{2}\}$

set of bonds $\mathcal{B}_L = \{ \{u, v\} \mid u \text{ and } v \text{ are nearest neighbors} \}$ (2)

bond $\{u, v\} = \{v, u\} \in \mathcal{B}_L$



$$|u - v| = 1$$

OR

$$\begin{cases} u = (u_1, \dots, \frac{L}{2}, \dots, u_d) \\ v = (u_1, \dots, -\frac{L}{2} + 1, \dots, u_d) \end{cases}$$

periodic boundary conditions

§ quantum spin systems on Λ_L

▷ single spin ($S=1/2$)

states $\alpha|\uparrow\rangle + \beta|\downarrow\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \in \mathbb{C}^2$ (1)

spin operators $\hat{S} = (\hat{S}^{(x)}, \hat{S}^{(y)}, \hat{S}^{(z)})$ (2)

$$\hat{S}^{(x)} = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \hat{S}^{(y)} = \frac{1}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \hat{S}^{(z)} = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (3)$$

▷ spin system

$\hat{S}_u^{(\alpha)}$ ($u \in \Lambda_L, \alpha = x, y, z$) copy of $\hat{S}^{(\alpha)}$ acting on $\mathcal{h}_u \cong \mathbb{C}^2$

total Hilbert space $\mathcal{H}_L = \bigotimes_{u \in \Lambda_L} \mathcal{h}_u$ (4)

basis
 $\{|\uparrow\rangle_u, |\downarrow\rangle_u\}$

basis $\{|\mathbb{T}\rangle\}_{\mathbb{T} \in \mathcal{S}}$ $|\mathbb{T}\rangle = \bigotimes_{u \in \Lambda_L} |\sigma_u\rangle_u$ (5)

$\mathcal{S} = \{\mathbb{T} = (\sigma_u)_{u \in \Lambda_L} \mid \sigma_u = \uparrow, \downarrow\}$ (6)

$\mathbb{T}$: spin configuration

► Total spin operators (order parameters)

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$$\hat{O}_L^{(\alpha)} = \sum_{u \in \Lambda_L} \hat{S}_u^{(\alpha)} \quad (1) \quad \alpha = x, y, z$$

θ -rotation about the α -axis $\hat{U}_L^{(\alpha, \theta)} = e^{-i\theta \hat{O}_L^{(\alpha)}} \quad (2)$

$$\left\{ \begin{aligned} (\hat{U}_L^{(z, \theta)})^\dagger \hat{O}_L^{(x)} \hat{U}_L^{(z, \theta)} &= \cos \theta \hat{O}_L^{(x)} - \sin \theta \hat{O}_L^{(y)} \end{aligned} \right. \quad (3)$$

$$\left\{ \begin{aligned} (\hat{U}_L^{(z, \theta)})^\dagger \hat{O}_L^{(y)} \hat{U}_L^{(z, \theta)} &= \sin \theta \hat{O}_L^{(x)} + \cos \theta \hat{O}_L^{(y)} \end{aligned} \right. \quad (4)$$

$$\left\{ \begin{aligned} (\hat{U}_L^{(z, \theta)})^\dagger \hat{O}_L^{(z)} \hat{U}_L^{(z, \theta)} &= \hat{O}_L^{(z)} \end{aligned} \right. \quad (5)$$

► translation operator

$$w \in \mathbb{Z}^d \quad \hat{T}_w |\mathbb{T}\rangle = |\mathbb{T}_{+w}\rangle \quad (6)$$

$$(\mathbb{T}_{+w})_u = \mathbb{T}_{u-w} \quad (7)$$

$\mathbb{T} \quad \dots \uparrow \downarrow \uparrow \uparrow \downarrow \uparrow \downarrow \uparrow \downarrow \uparrow \downarrow \downarrow \downarrow \uparrow \downarrow \uparrow \dots$
 $\mathbb{T}_{+2} \quad \dots \downarrow \downarrow \uparrow \downarrow \uparrow \uparrow \downarrow \uparrow \downarrow \uparrow \downarrow \uparrow \downarrow \downarrow \downarrow \uparrow \dots$